Fixed program responses with Reactome-only annotation
row ordering is display-only and does not alter Louvain membership

Reactome-labelled program-response atlas

Observed 24 h | Model 24 h | Observed 1 week | Model 1 week

Genes 3–12 | Seed Jaccard | Reactome ORA label

P311 — Neutrophil Degranulation
P137 — Cell Cycle Checkpoints
P288 — Sensory Perception
P304 — Extracellular Matrix Organization
P023 — Cell Cycle
P045 — Cell Cycle

All 393 fixed Louvain embedding programs
(only provisional Reactome labels shown)

−0.6 −0.4 −0.2 0.0 0.2 0.4 0.6

Raw gene-program score difference (red = increased; blue = decreased)

○ Within-run program-effect FDR < 0.05 ■ Provisional Reactome within-program ORA label